THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHS · CLASS 9 · SET 03

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Objective Type — 20 questions · 20 marks

Q1–Q18 are multiple-choice questions (1 mark each). Q19–Q20 are Assertion (A) and Reason (R) questions: select the correct option from (a), (b), (c), (d) given below each.

1. Which of the following points lies in the second quadrant?

(a) (−3, −4) (b) (3, 4) (c) (−3, 4) (d) (3, −4)

Ans: (c) Second quadrant: $x < 0$ and $y > 0$.

2. The distance of the point (−8, 15) from the origin is:

(a) 7 units (b) 13 units (c) 23 units (d) 17 units

Ans: (d) $d = \sqrt{(64 + 225)} = \sqrt{289} = 17$.

3. The graph of the equation $x + 2y = 6$ cuts the y-axis at the point:

(a) (6, 0) (b) (0, 6) (c) (0, 3) (d) (3, 0)

Ans: (c) Put $x = 0$: $2y = 6 \Rightarrow y = 3$.

4. Which of the following is a rational number?

(a) $\sqrt{7}$ (b) $\sqrt{(121/9)}$ (c) π (d) $\sqrt{3} + 1$

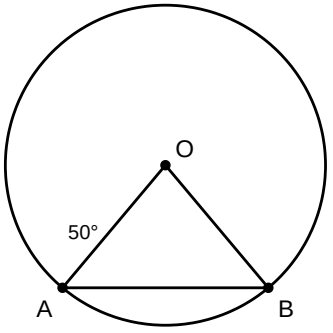
Ans: (b) $\sqrt{(121/9)} = 11/3$ is rational; $\sqrt{7}$, π and $\sqrt{3} + 1$ are irrational.

5. If $a + b = 7$ and $ab = 10$, then $a^2 + b^2$ is equal to:

(a) 24 (b) 49 (c) 39 (d) 29

Ans: (d) $a^2 + b^2 = (a + b)^2 - 2ab = 49 - 20 = 29$.

6. In the figure, O is the centre of the circle and $\angle OAB = 50°$. Then $\angle AOB$ is equal to:



(a) 100° (b) 80° (c) 130° (d) 50°

Ans: (b) $OA = OB \Rightarrow \angle OBA = 50°$; $\angle AOB = 180° - 100° = 80°$.

7. Two chords of a circle which are at the same distance from the centre are:

(a) equal in length (b) parallel

(c) perpendicular to each other (d) diameters

Ans: (a) Chords equidistant from the centre are equal.

8. The area of a triangle with sides 6 cm, 8 cm and 10 cm, found using Heron's formula, is:

(a) 40 cm² (b) 30 cm² (c) 48 cm² (d) 24 cm²

Ans: (d) s = 12; area = $\sqrt{(12 \times 6 \times 4 \times 2)} = \sqrt{576} = 24$ cm².

9. A fair die is thrown once. The probability of getting a multiple of 3 is:

(a) $\frac{2}{3}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

Ans: (c) Favourable outcomes 3 and 6 ⇒ 2/6 = 1/3.

10. If $t_n = 2^n$, then t_6 is:

(a) 12 (b) 32 (c) 64 (d) 36

Ans: (c) $2^6 = 64$.

11. The converse of 'If n is the square of a prime number, then n has exactly 3 factors' is:

(a) If n is a prime number, then n has exactly 3 factors.

(b) If n has exactly 3 factors, then n is the square of a prime number.

(c) If n has exactly 3 factors, then n is a prime number.

(d) If n is not the square of a prime, then n does not have exactly 3 factors.

Ans: (b) Interchange the 'if' and 'then' parts.

12. A counterexample to the claim '$4^n + 3$ is prime for every positive integer n' is:

(a) n = 2 (b) n = 4 (c) n = 3 (d) n = 1

Ans: (b) $4^4 + 3 = 259 = 7 \times 37$. (n = 1, 2, 3 give 7, 19, 67, all prime.)

13. A student scores 50, 60 and 80 in three components which carry weights in the ratio 1 : 2 : 7. The weighted mean score is:

(a) 70 (b) 75 (c) 73 (d) 63.3

Ans: (c) $(50 \times 1 + 60 \times 2 + 80 \times 7)/(1 + 2 + 7) = 730/10 = 73$.

14. The surface area of a sphere of radius 7 cm ($\pi = 22/7$) is:

(a) 308 cm² (b) 154 cm² (c) 616 cm² (d) 1437 cm²

Ans: (c) $4\pi r^2 = 4 \times 22/7 \times 49 = 616$ cm².

15. To find gcd(m, n) by searching for common divisors, it is enough to check numbers up to:

(a) min(m, n) (b) max(m, n) (c) m + n (d) m × n

Ans: (a) A common divisor cannot be larger than the smaller of the two numbers.

16. In a parallelogram ABCD, AB = (2x + 3) cm and CD = (5x − 6) cm. Then the length of AB is:

(a) 15 cm (b) 9 cm (c) 7 cm (d) 6 cm

Ans: (b) Opposite sides are equal: 2x + 3 = 5x − 6 ⇒ x = 3 ⇒ AB = 9 cm.

17. The diagonals of a quadrilateral ABCD are 12 cm and 18 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is:

(a) 60 cm (b) 15 cm (c) 45 cm (d) 30 cm

Ans: (d) Sides of the Varignon parallelogram are half the diagonals: 6 cm and 9 cm ⇒ perimeter = 2(6 + 9) = 30 cm.

18. The sum of the first five terms of the AP 3, 7, 11, … is:

(a) 50 (b) 55 (c) 45 (d) 65

Ans: (b) $S_5 = 5/2 \times [2(3) + 4(4)] = 5/2 \times 22 = 55$.

19. **Assertion (A):** The probability of an event can be 1.2.
Reason (R): The probability of any event lies between 0 and 1, both inclusive.

Ans: (d) A is false because probability never exceeds 1; R is a true statement.

20. **Assertion (A):** A quadrilateral in which one pair of opposite sides is equal and parallel is a parallelogram.
Reason (R): In such a quadrilateral the diagonals bisect each other.

Ans: (a) Both true: the congruent triangles formed by the diagonals show that they bisect each other, which makes the quadrilateral a parallelogram.

SECTION B — Very Short Answer (2 marks each) — 5 questions · 10 marks

Internal choice is provided in one question.

21. An internet café charges ₹20 for the first hour and ₹12 for every additional hour. Write an expression for the charge for h hours (h ≥ 1) and find the charge for 5 hours.

Charge = 20 + 12(h − 1) (1)
For h = 5: 20 + 12 × 4 = **₹68** (1)

22. Without actual division, decide whether $\frac{13}{60}$ has a terminating decimal expansion. Give a reason.

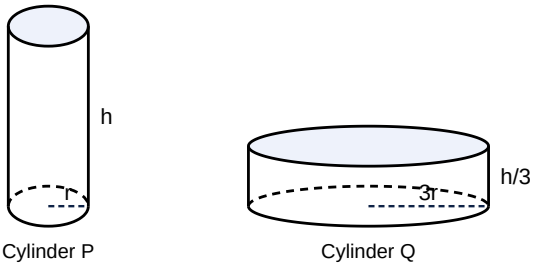
60 = $2^2 \times 3 \times 5$ (1). The prime factor 3 is present, so the decimal expansion is **non-terminating (repeating)** (1)

O R

Find two rational numbers between 0.75 and 0.8.

0.75 = 0.750 and 0.8 = 0.800 (1). Two numbers: **0.76 and 0.77** (any two correct numbers) (1)

23. The radius of cylinder Q is three times the radius of cylinder P and its height is one-third of the height of P. Find the ratio of (i) their curved surface areas, (ii) their volumes (P : Q).

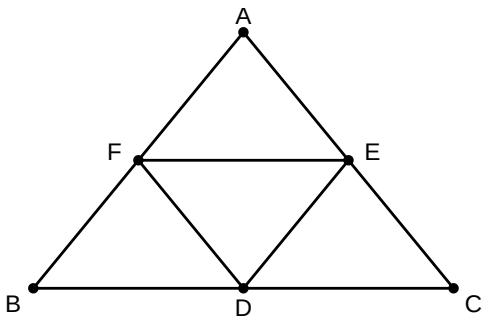


Let P have r, h; Q has 3r, h/3. (i) CSA: 2πrh : 2π(3r)(h/3) = **1 : 1** (1)
(ii) Volume: $\pi r^2 h : \pi(9r^2)(h/3) = \pi r^2 h : 3\pi r^2 h$ = **1 : 3** (1)

24. Execute the ‘divisors’ algorithm (check every j from 1 to n and add j to the list if it divides n) for n = 24 and write the final list.

j = 1, 2, 3, 4, 6, 8, 12 and 24 divide 24; the others do not (1)
List = **[1, 2, 3, 4, 6, 8, 12, 24]** (1)

25. In the figure, D, E and F are the midpoints of BC, CA and AB. If AB = 12 cm, BC = 16 cm and CA = 10 cm, find the perimeter of △DEF.



EF = ½BC = 8, DE = ½AB = 6, DF = ½CA = 5 (Midpoint Theorem) (1)
Perimeter = 8 + 6 + 5 = **19 cm** (1)

SECTION C — Short Answer (3 marks each) — 6 questions · 18 marks

Internal choice is provided in two questions.

26. Express 3.1666… (6 repeating) in the form p/q in lowest terms.

x = 3.1666…; 10x = 31.666…, 100x = 316.666… (1)
100x − 10x: 90x = 285 (1)
x = 285/90 = **19/6** (1)

27. A box contains 3 red, 4 blue and 5 yellow marbles. One marble is drawn at random. Find the probability that it is (i) blue, (ii) not yellow, (iii) red or blue.

Total = 12. (i) 4/12 = **1/3** (1) (ii) 7/12 (1) (iii) (3 + 4)/12 = **7/12** (1)

OR

A bus ran late on some of the 50 days of a month. The minutes it was late are recorded:

Minutes late	0	1	2	3
Days	18	20	8	4

Find the probability that on a day chosen at random the bus was (i) on time, (ii) at least 2 minutes late, (iii) at most 1 minute late.

Total = 50. (i) 18/50 = **9/25** (1) (ii) (8 + 4)/50 = **6/25** (1) (iii) (18 + 20)/50 = **19/25** (1)

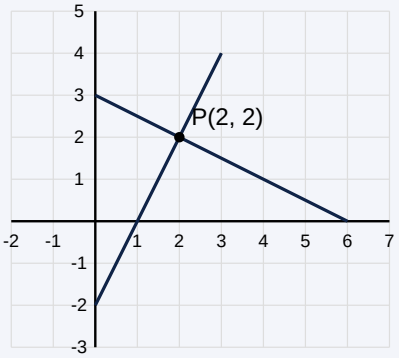
28. Draw the graphs of the equations x + 2y = 6 and 2x − y = 2 on the same grid and use them to find the solution of the pair.

Table for x + 2y = 6:

x	0	6
y	3	0

Table for 2x − y = 2:

x	0	1
y	-2	0



(1) Draw both lines (1) The lines meet at **(2, 2)**, so x = 2, y = 2 (1)

29. The monthly expenses of a family are: food ₹6000, rent ₹4500, education ₹3000 and others ₹1500. Find the total expense and the percentage share of each item in a 100% stacked bar.

Total = ₹15000 (1)
Food = 6000/15000 × 100 = **40%**; rent = **30%**; education = **20%**; others = **10%** (2: ½ each + check that the shares add up to 100%)

30. A chord AB of length 16 cm and a chord CD of length 12 cm are drawn in a circle of radius 10 cm. Find their distances from the centre and show that the longer chord is nearer the centre.

d(AB): half = 8 ⇒ d = √(100 − 64) = 6 cm (1)
d(CD): half = 6 ⇒ d = √(100 − 36) = 8 cm (1)
6 cm < 8 cm, so the longer chord AB is nearer the centre (by 2 cm) (1)

OR

Using Heron's formula, find the area of a triangle with sides 13 cm, 20 cm and 21 cm.

s = (13 + 20 + 21)/2 = 27 cm (1)
Area = √[27 × 14 × 7 × 6] (1) = √15876 = **126 cm²** (1)

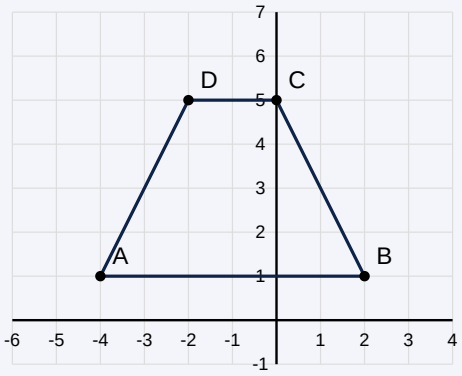
31. Prove that the opposite angles of a parallelogram are equal.

Let ABCD be a parallelogram, so AB ∥ DC and AD ∥ BC. (½)
AD ∥ BC with transversal AB: ∠A + ∠B = 180° (co-interior angles) (1)
AB ∥ DC with transversal BC: ∠B + ∠C = 180° (1)
Hence ∠A = ∠C. In the same way ∠B = ∠D. (½)

SECTION D — Long Answer (5 marks each) — 4 questions · 20 marks

Internal choice is provided in two questions.

32. A shopkeeper draws the plan of a stall on a grid with corners A(−4, 1), B(2, 1), C(0, 5) and D(−2, 5).
(a) Plot the points and join them in order. (2 marks)
(b) Show that AB ∥ DC and that AD = BC. (1½ marks)
(c) Name the quadrilateral and find its area. (1½ marks)



(a) Plotting 4 points (1½), joining (½): (b) AB and DC are both horizontal (same y-coordinate) ⇒ AB ∥ DC. AD = √(2² + 4²) = √20; BC = √(2² + 4²) = √20 ⇒ AD = BC (1½)
(c) One pair of parallel sides and equal non-parallel sides: **isosceles trapezium** (½). Parallel sides 6 and 2, height 4 ⇒ area = ½(6 + 2) × 4 = **16 square units** (1)

33. How many multiples of 6 lie between 100 and 600? Find their sum. (2½ + 2½)

Multiples: 102, 108, …, 594 — an AP with a = 102, d = 6, l = 594 (1)
n = (594 − 102)/6 + 1 = **83** (1½)
Sum = 83/2 × (102 + 594) = 83 × 348 = **28884** (2½)

OR

A traffic survey recorded the vehicles passing a crossing in an hour:

Vehicle	Two-wheeler	Car	Bus	Truck
Number	240	120	60	80

A vehicle is chosen at random. Find the probability that it is (i) a two-wheeler (1), (ii) not a car (2), (iii) a bus or a truck (2).

Total = 500. (i) 240/500 = **12/25** (1) (ii) 1 − 120/500 = 380/500 = **19/25** (2) (iii) (60 + 80)/500 = **7/25** (2)

34. State and prove the Baudhāyana–Pythagoras converse: if a² + b² = c² for the sides of a triangle, then the triangle is right-angled.

Let △ABC have BC = a, CA = b, AB = c with $a^2 + b^2 = c^2$. (1)
Draw △XYZ with ∠Z = 90°, YZ = a, XZ = b. By Pythagoras $XY^2 = a^2 + b^2 = c^2 \Rightarrow XY = c$ (2)
△ABC ≅ △XYZ by SSS (1) ⇒ ∠C = ∠Z = 90° (½)
So **△ABC is right-angled at C**. (½)

OR

A goldsmith melts 9 units of gold of purity 12 varṇa, 5 units of purity 10 varṇa and x units of purity 14 varṇa. The mixture has purity 12 varṇa. Find x.

Purity of mixture = (9 × 12 + 5 × 10 + x × 14)/(9 + 5 + x) (2)
12 = (108 + 50 + 14x)/(14 + x) ⇒ 168 + 12x = 158 + 14x (2)
2x = 10 ⇒ **x = 5 units** (1)

35. The sum of the length and the breadth of a rectangular plot is 17 m and its area is 72 m². Form a quadratic expression in the breadth x, factorise it and find the dimensions.

Length = 17 − x; $x(17 - x) = 72 \Rightarrow x^2 - 17x + 72 = 0$ (1½)
(x − 8)(x − 9) = 0 (1½)
x = 8 or 9 (1). The plot is **9 m by 8 m** (1)

OR

The area of a rectangular carpet is $(6x^2 + 17x + 12)$ m² and its breadth is (2x + 3) m. Find its length and, for x = 2, its perimeter.

$6x^2 + 17x + 12 = 6x^2 + 9x + 8x + 12$ (1½) $= 3x(2x + 3) + 4(2x + 3) = (2x + 3)(3x + 4)$ (1)
Length = **(3x + 4) m** (1)
For x = 2: breadth 7 m, length 10 m ⇒ perimeter = 2(7 + 10) = **34 m** (1½)

SECTION E — Case-Study Based Questions (4 marks each) — 3 questions · 12 marks

Read each case carefully and answer the questions that follow. Internal choice is provided in every 2-mark sub-part.

36. A mobile shop sold items to 120 customers in a week: mobile phones 45, laptops 30, tablets 25 and accessories 20.

(i) Find the probability that a customer chosen at random bought a laptop.

30/120 = **1/4**

(ii) Find the probability that the customer did not buy an accessory.

1 − 20/120 = 100/120 = **5/6**

(iii) Find the probability that the customer bought a mobile phone or a tablet.

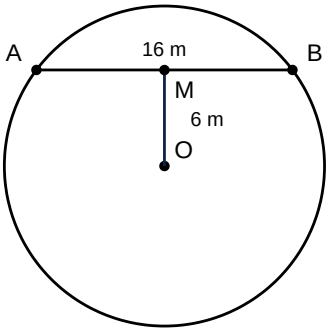
(45 + 25)/120 (1) = 70/120 = **7/12** (1)

OR

(iii) Another 30 customers bought laptops. Find the new probability that a customer chosen from all customers bought a laptop.

Laptops = 60, total = 150 (1) ⇒ 60/150 = **2/5** (1)

37. A circular pond has centre O and radius 10 m. A fence AB, 16 m long, is a chord of the pond. M is the midpoint of AB and OM = 6 m.



(i) Find the length AM.

OM ⊥ AB bisects the chord ⇒ AM = 16/2 = **8 m**

(ii) Verify that the radius of the pond is 10 m.

$OA^2 = OM^2 + AM^2 = 36 + 64 = 100 \Rightarrow OA =$ **10 m**

(iii) A second fence CD is placed at a distance of 8 m from O. Find its length.

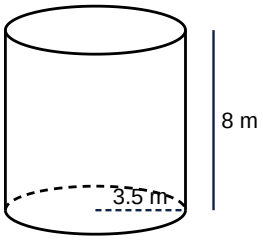
Half of CD = √(100 − 64) = 6 (1) ⇒ CD = **12 m** (1)

OR

(iii) Find the area of the triangle OAB.

Area = ½ × AB × OM = ½ × 16 × 6 (1) = **48 m²** (1)

38. A school has an open cylindrical water tank of radius 3.5 m and height 8 m (π = 22/7).



Water tank

(i) Find the curved surface area of the tank.

CSA = 2πrh = 2 × 22/7 × 3.5 × 8 = **176 m²**

(ii) Find the volume of water the tank can hold.

$V = \pi r^2 h = 22/7 \times 12.25 \times 8 =$ **308 m³**

(iii) Express the capacity of the tank in litres (1 m³ = 1000 litres).

308 × 1000 (1) = **308000 litres** (1)

OR

(iii) The outer curved surface is painted at ₹40 per m². Find the cost.

Cost = 176 × 40 (1) = **₹7040** (1)